PHISHING DETECTION & AWARENESS REPORT- Task 2

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1. Introduction

Phishing remains one of the most widespread and effective cyber threats targeting both individuals and organisations. It involves attackers impersonating trusted entities to manipulate users into revealing sensitive information such as passwords, banking details, or one-time PINs (OTPs).

Unlike many cyberattacks that exploit system vulnerabilities, phishing relies on human behaviour. Attackers craft messages that appear urgent, convincing, or beneficial, making it easy for users to act without fully verifying the source.

2. Objective

The purpose of this report is to analyse phishing email samples in order to understand how these attacks are structured and executed. The report also aims to highlight common warning signs and provide practical guidance to help users recognise and avoid phishing attempts.

3. Methodology

The following approach was used to carry out the analysis:

- Collected representative phishing email samples
- Examined the tone, structure, and wording of each message
- Inspected embedded links and sender details
- Identified recurring phishing indicators
- Classified each email based on its risk level
- Summarised findings in a clear and accessible format

Basic inspection techniques were applied, including reviewing link structures and identifying inconsistencies in messaging. Public email analysis tools were referenced where applicable to support the evaluation.

4. Email Analysis

Email 1: Account Verification Scam

Subject: ⚠️ Urgent: Your Account Will Be Locked

Email Body:

Dear User,
We noticed suspicious activity on your account.
To avoid account suspension, please verify your details immediately.

Verify Now: http://secure-account-verify[.]com

Failure to verify within 24 hours will result in permanent account lock.

Regards,
Security Team

Phishing Indicators

- The message uses a **generic greeting**, suggesting it was sent in bulk
- It creates **urgency and fear** by threatening account suspension
- The **link provided appears untrusted** and does not match a known service
- There is **no clear company identity or branding**, reducing credibility

Risk Classification:

PHISHING (High Risk)

Analysis

This message is designed to pressure the recipient into acting quickly without verification. The use of urgency is a common tactic to bypass critical thinking. In legitimate scenarios, service providers typically allow users to verify account activity through official platforms rather than external links. The structure and tone strongly indicate an attempt to redirect the user to a fraudulent website to capture login credentials.

Email 2: Delivery Issue Email (SUSPICIOUS)

Subject: Delivery Attempt Failed – Action Required

Email Body:

Dear Customer,

We attempted to deliver your package but were unsuccessful.
Please reschedule your delivery by clicking the link below:
http://delivery-reschedule-service[.]com

Thank you,
Delivery Team

Indicators

- **Generic greeting** ("Dear Customer")
- **Unexpected message** (user may not be expecting a package)
- Contains a **clickable link**
- Sender is **vague and unverified**

Risk Classification:

SUSPICIOUS (Medium Risk)

Analysis

This email could be legitimate, but there are several warning signs. The lack of personalisation and the presence of an unverified link raise concerns. Without confirming whether a delivery was expected or verifying the sender, it is unsafe to interact with the email. Further validation is required before taking action.

Email 3: Gift Card Scam

Subject: 🎁 You've Won a Gift Card!

Email Body:

Dear Customer,

Congratulations! You have been selected to receive a $500 gift card.

To claim your reward, click the link below:
http://free-gift-claim-now[.]com

Hurry, offer expires today!

Best regards,
Rewards Team

Phishing Indicators

- The offer is **unexpected and unrealistic**, which is a common lure
- The message uses **time pressure** to rush the user's decision
- The **link does not correspond to any legitimate brand**
- The sender identity is **vague and cannot be verified**

Risk Classification:

PHISHING (High Risk)

Analysis

This type of attack relies on curiosity and excitement rather than fear. By presenting an attractive reward, the attacker increases the likelihood of user interaction. Once the link is accessed, the user may be directed to a fake website designed to collect personal or financial information. Legitimate organisations do not distribute rewards in this manner without prior engagement or verification.

Email 4: Payroll Update Scam

Subject: Payroll Update Required

Email Body:

Dear Employee,

We are updating our payroll system. Please confirm your banking details to avoid payment delays.

Click here to update:
http://company-payroll-update[.]net

Regards,
HR Department

Phishing Indicators

- The email requests **highly sensitive financial information**
- A **generic greeting** indicates lack of personalisation
- The **domain used in the link is suspicious and unofficial**
- The attacker **impersonates an internal department (HR)** to build trust

Risk Classification:

PHISHING (High Risk)

Analysis

This message targets employees by exploiting trust in internal communication channels. Because payroll is critical, users may feel obligated to respond quickly. However, legitimate organisations typically handle such updates through secure internal systems rather than external links. The request for banking details is a strong indicator of a phishing attempt aimed at financial fraud.

Email 5: Legitimate Notification

Subject: Your Monthly Bank Statement is Ready

Email Body:

Dear Samantha Swart,

Your monthly bank statement is now available.
Please log in to your official banking app or website to view your statement securely.

For your security, we do not include links in this email.

Kind regards,
ABC Bank

Indicators

- Uses **your actual name** (personalised)
- **No suspicious links** included
- Encourages using **official app/website**
- Clear and professional tone
- Identifiable organisation

Risk Classification:

SAFE (Low Risk)

Analysis

This email follows good security practices by avoiding clickable links and directing the user to access their account through official channels. The tone is calm and informative, with no urgency or pressure, which is typical of legitimate communication.

Email 6: Account Security Alert (CLEAR PHISHING)

Subject: Important: Unusual Login Attempt Detected

Email Body:

Dear User,

We detected an unusual login attempt from a new device.
If this was not you, please secure your account immediately:
http://account-security-check[.]info

Failure to act may result in account suspension.

Security Team

Indicators

- Generic greeting
- Fear-based language (unusual login + suspension)
- Suspicious domain (.info)
- Encourages immediate action via link

Risk Classification:

PHISHING (High Risk)

<u>Analysis</u>

This email mimics legitimate security alerts but uses urgency and fear to manipulate the user. The link provided is not associated with a trusted domain, indicating a likely phishing attempt designed to steal login credentials.

6. Common Phishing Techniques Identified

The analysis reveals several recurring techniques used by attackers:

- **Urgency and pressure** to force immediate action
- Use of **malicious or misleading links**
- **Generic greetings** to target multiple users
- **Impersonation of trusted entities** (e.g., HR, security teams)
- Use of **incentives or threats** to manipulate behaviour

These techniques are designed to reduce user suspicion and encourage quick, unverified responses.

7. Prevention & Awareness Guidelines

Recommended Practices

- Carefully verify the sender's email address before responding
- Hover over links to check their actual destination
- Confirm unusual requests through official communication channels
- Report suspicious emails to the relevant IT or security team
- Use multi-factor authentication (MFA) for added security

Practices to Avoid

- Avoid clicking on links from unknown or suspicious emails
- Never share passwords, banking details, or OTPs via email
- Do not trust messages that create unnecessary urgency
- Avoid downloading unexpected attachments
- Do not assume an email is safe based solely on appearance

8. Conclusion

The emails analysed in this report demonstrate how phishing attacks are carefully designed to manipulate user behaviour through urgency, trust, and deception. Although the messages may appear convincing at first glance, closer inspection reveals clear warning signs.

A key observation is that these emails closely resemble everyday communication, which increases their effectiveness. This highlights the importance of user awareness as a primary line of defence.

By understanding common phishing techniques and applying basic verification practices, users can significantly reduce the risk of falling victim to such attacks. Ultimately, awareness and caution remain the most effective tools in preventing phishing-related incidents.